When the Regime Returns: Retention and Invariance in Decision-Focused Strategy Pools for Non-Stationary Markets

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Abstract

Financial markets are non-stationary in two distinct ways that existing strategy-allocation methods conflate. **Between regimes**, market states (trend, range, high-volatility, crisis) switch and recur on irregular schedules with long dormancy, so a reward-chasing capital allocator—recent-P&L weighting or a learned mixture-of-experts router—starves and overwrites the dormant-regime specialist and pays a full recalibration cost exactly when the regime returns. **Within a regime**, each recurrence is a different draw from the regime's environment distribution (the 2008 crisis is not 2020 is not 2022), so a specialist trained by empirical risk minimization on past episodes of its own regime still fails out-of-distribution on the next one. We show these failure modes are governed by *independent* design choices and propose RISP (Regime-Invariant Specialist Pools), a pool of capacity-bounded, decision-focused specialists in which (i) regimes are acquired by batched winner-take-all competition, yielding a converged regime-to-specialist assignment that is *independent of current reward*—the crisis specialist idles through calm markets and retains its calibration by identity, not P&L—and (ii) each specialist trains its niche with a decision-focused *invariance* objective (variance plus mean of an SPO-style regret surrogate across historical episodes of its regime). We prove a post-reactivation regret bound that decomposes into a *forgetting deficit*—zero for any reward-independent assignment, approaching the full relearning cost for any reward-driven one—and an *invariance gap* controlled by the episode-variance penalty, including the Pinsker-type bridge from divergence-metric invariance to regret. In controlled regime-switching markets with realistic signal-to-noise, the predicted 2×2 dissociation holds with all pre-registered orderings confirmed (20 seeds, Holm-corrected): emergent reward-independent assignment cuts post-reactivation decision regret by 6.1% versus a capacity-matched monolith (p=1.3×10^{-3}), the invariance objective cuts it a further 4.4% (p=6.1×10^{-3}; 10.3% jointly, p=9.2×10^{-7}; 38% of the excess over the irreducible noise floor), and the interaction is *super-additive*: invariant training buys nothing in a reward-driven monolith (p=0.88 vs. ERM), because an objective cannot help a head that has been evicted. The learned router and the recent-performance allocator inherit the monolith's failure; the full system is statistically indistinguishable from a hand-pinned oracle assignment trained with the same objective (p=0.72)—recovering the skyline without being handed the assignment—and nominally beats the oracle assignment trained with ERM. We pair the mechanism with an honest audit in the GAUSE tradition: a pre-registered structure diagnostic run on real crypto and commodity panels finds *no exploitable regime structure on the decision-regret metric* under causal, leakage-free labelings—so the real-data 2×2 is correctly null, the mechanism's preconditions are checkable before deployment, and we quantify exactly when retention is worthless (high signal-to-noise relearning, short dormancy, structureless regimes). All claims are regret and retention claims, not alpha claims; code, data, and per-figure result files are released.

1 Introduction

A multi-strategy fund maintains playbooks for market conditions that are mostly absent: the crisis desk idles through years of calm; the short-volatility book hibernates through stress. The allocation question that governs such a system—*which strategy gets capital, attention, and recalibration today, and who keeps a strategy whose regime is dormant?*—has a standard machine-learning answer (route by recent reward: a learned gate, a recent-Sharpe weighting) and a standard failure mode that the GAUSE companion study isolated in controlled prediction domains (Li, 2026): a dormant regime emits no reward, so any allocator that protects capacity using current reward is structurally blind to exactly the knowledge that most needs protection. When the regime returns, the system relearns what it once knew, and in markets the relearning window—the first days of a crisis—is precisely when decisions are most consequential.

This paper develops that observation into a theory and system for *decision-focused* learning in non-stationary markets, and in doing so isolates a second, orthogonal failure mode that the retention story alone misses. Suppose the crisis specialist *is* protected—its parameters untouched through dormancy. Protection guarantees only that the specialist still knows *the last crisis*. Each occurrence of a regime is a distinct draw from that regime's environment distribution: flight-to-quality correlations, volatility clustering, and liquidity spirals recur *structurally*, but the surface features that carried them in 2008 differ from 2020 and from 2022. A specialist trained by empirical risk minimization (ERM) on its own regime's history loads on episode-specific ("spurious") regularities and fails out-of-distribution on the next episode (Arjovsky et al., 2019; Yan et al., 2026). Retention solves *between-regime* forgetting; it does nothing for *within-regime* drift.

Our central claim is that these are **two orthogonal kinds of non-stationarity**, addressed by *independent* mechanisms:

- **(N1) Between-regime switching** is an *allocation* problem. It is solved by any capacity assignment that is *independent of current reward* and persistently covers the regime inventory—so that the dormant specialist receives no foreign updates and retains by identity. No training objective can substitute: a forgotten head is gone regardless of the loss it was trained with.
- **(N2) Within-regime drift** is a *training-objective* problem. It is solved by an invariance objective across the regime's historical episodes—penalizing the variance of a decision-focused (SPO-style) regret surrogate across episodes so the specialist learns the decision-relevant structure common to *all* crises rather than the idiosyncrasies of the last one. No allocation mechanism can substitute: retention of an episode-overfit head retains the overfitting too.

The paper's single falsifiable claim is the resulting 2×2: crossing allocation mechanism (reward-driven vs. reward-independent) with training objective (ERM vs. invariant) should produce monotone improvement along both axes in post-reactivation decision regret, with a signature asymmetry—the invariance axis should be inert in the reward-driven row, because an evicted head cannot benefit from the objective that trained it. Our experiments confirm every cell of this prediction, including the asymmetry (Section 5.2), and our theory derives it (Section 4).

Scope statement (the sleeping-experts deflation, addressed up front). The contribution concerns capacity-bounded *learners*, not combinations of fixed experts. Sleeping-experts and adaptive-regret schemes (Freund et al., 1997; Hazan and Seshadhri, 2009; Daniely et al., 2015) also freeze dormant experts' state—but only because their experts are *static strategies* with nothing to forget. The problem RISP solves arises precisely when specialists must keep *learning* within their niche under bounded capacity, so that reward-driven reallocation causes destructive interference across regimes. We operationalize this distinction as a baseline pair: Hedge over *fixed* strategies retains trivially but cannot adapt within-regime and is dominated by every learning arm (Section 5.2); Hedge over capacity-bounded *learning* experts inherits the interference failure and tracks the monolith. The most dangerous related work thus becomes a supporting experiment.

Contributions.

1. **Problem decomposition (C1).** We formally decompose non-stationary financial decision error into a between-regime *forgetting deficit* and a within-regime *invariance gap*, and show—in theory and

experiment—that they are governed by independent design choices (allocation vs. training objective), with a predicted and confirmed super-additive interaction: invariant training pays *only* inside a retaining allocation.

2. **Method (C2).** RISP: the first strategy-pool architecture combining reward-independent, emergent (competition-driven) capacity assignment with decision-focused invariant learning per niche. The anticipated financial adaptation of the GAUSE competition mechanism is a *batched* winner-take-all window (W_c days of cumulative within-regime regret); we pre-registered the expectation that per-step competition would converge to noise at realistic signal-to-noise, and report honestly that the expectation is *not* borne out—the exponentiated-gradient affinity accumulation across windows performs the necessary averaging, and the mechanism is robust across $W_c \in \{1, \ldots, 60\}$ (Section 5.8). No gate is trained, no freezing schedule is set, no regime-to-desk map is designed.

3. **Theory (C3).** A post-reactivation regret bound $\mathbb{E}[\rho_{\text{react}}] \leq G_{\text{inv}} + G_{\text{forget}}$ (Proposition 2) in which $G_{\text{forget}} = 0$ for *any* reward-independent covering assignment while reward-driven assignment drives it to the full relearning cost at an explicit rate (Proposition 1); G_{inv} is controlled by the episode-variance penalty (Lemma 1), with an explicit Pinsker-type bridge from divergence-metric invariance statements to the regret metric (Lemma 2)—closing the technical gap between Inv-PnCO-style guarantees (Yan et al., 2026) and decision regret. A break-even analysis (Proposition 4) gives the dormancy window in which retention pays once idle-specialist carrying costs are charged.

4. **Evaluation protocol (C4).** A diagnostic-first standard for regime-aware financial ML: *post-reactivation regret* in a probe window after each reactivation, leakage-free causal regime labelings, and a pre-registered structure diagnostic run *before* claiming gains. On our real panels (crypto, commodities) the diagnostic correctly *fails*—no exploitable regime structure on the decision metric under causal labels—and the real-data dissociation is, as the methodology then requires, null. We report this prominently rather than burying it: where the diagnostic fails, no regime-aware mechanism (ours included) can be bought, and a method that had claimed otherwise would be fitting noise.

5. **Honest audit (C5).** Conditions under which RISP does *not* pay, each quantified: slack capacity ($K \to R$ closes the monolith gap); high within-regime signal-to-noise (a fresh learner recalibrates within days, making retention worthless—we sweep the break-even relearning speed); short dormancy (the retention gap grows with dormancy length); structureless regimes (the E0 gate); and carrying costs above the break-even threshold.

We emphasize register throughout: our claims are *regret and retention* claims about a mechanism, demonstrated where its preconditions verifiably hold, not alpha claims about live markets. The honest result on present real data is that the preconditions do *not* hold there at daily frequency under our labelers and feature inventory—a finding the diagnostic-first protocol is designed to surface, and one consistent with the GAUSE oracle audit on the prediction metric (Li, 2026).

2 Related Work

Regime-switching finance. Markov-switching models (Hamilton, 1989), regime effects in asset returns (Ang and Timmermann, 2012), regime-conditioned allocation (Guidolin and Timmermann, 2007; Nystrup et al., 2018), and statistical jump models for persistent market states (Nystrup et al., 2020) detect regimes and condition a *single* model on them. None address bounded capacity or dormancy-driven forgetting of the specialist itself: the implicit assumption is that conditioning information is free to retain. Our monolith arm (regime-conditioned, capacity-bounded, always trained on the active regime) is this literature's best case under a capacity budget, and the comparison isolates what conditioning alone cannot provide.

Decision-focused learning (predict-then-optimize). The SPO framework and its convex surrogate (Elmachtoub and Grigas, 2022), surrogate relaxations for hard combinatorial problems (Mandi et al., 2020), learned decision losses (Shah et al., 2022), and the field survey (Mandi et al., 2024) all assume a stationary (or one-shot OOD) environment. Inv-PnCO (Yan et al., 2026) introduces invariant learning for predict-and-optimize under distribution shift, bounding the divergence between solution distributions across environments.

We adapt its environment-indexed invariance to *recurring* regimes with dormancy—episodes of a regime as environments—and contribute the bridge from divergence-metric guarantees to regret (Lemma 2), plus the orthogonal allocation axis that the stationary setting never encounters.

Online learning and ensembles. Hedge/multiplicative weights (Freund and Schapire, 1997; Arora et al., 2012), universal portfolios (Cover, 1991), online portfolio selection (Li and Hoi, 2014), and the sleeping-experts / adaptive-regret line (Freund et al., 1997; Hazan and Seshadhri, 2009; Daniely et al., 2015) are the closest formal relatives. The scope statement of Section 1 is our differentiation: these schemes combine *fixed* experts, for which dormancy is harmless by construction. When the experts are capacity-bounded *learners*, the combination scheme's weights are reward-driven and the training resource follows the weights—reintroducing exactly the interference that retention requires avoiding. Our A8a/A8b baseline pair tests both readings directly.

Mixture-of-experts and continual learning. MoE theory for continual learning (Li et al., 2025) shows isolation mitigates forgetting *provided* experts are plentiful and the gate's updates are eventually terminated. GAUSE (Li, 2026) showed empirically that in the excluded regime (bounded capacity, gate still learning) a reward-driven router inherits the monolith's forgetting, and that emergent competitive exclusion reaches the required reward-independent assignment with no freezing schedule. We import that mechanism as a premise, port it to decision-focused financial learning (batched competition, regret-based quality), and extend the theory from prediction error to decision regret. Classical continual-learning remedies (EWC (Kirkpatrick et al., 2017), progressive nets (Rusu et al., 2016); survey (Parisi et al., 2019)) operate within a single model and are complementary.

Multi-strategy practice and strategy decay. The practitioner analogue of our mechanism is the siloed desk structure of multi-strategy funds; the practitioner analogue of our failure mode is recent-P&L capital allocation *across* desks. Strategy decay (McLean and Pontiff, 2016) and backtest-overfitting critiques (Bailey and López de Prado, 2014; Harvey and Liu, 2016; López de Prado, 2018) motivate both our within-regime drift axis and our statistical protocol (deflated statistics, no test-period tuning, controlled semi-synthetic ground truth).

3 Problem Setup and Method

3.1 Data stream, episodes, and the decision layer

A stream $t = 1, \ldots, T$ presents observable per-asset features $X_t \in \mathbb{R}^{n\times d}$ and an unknown coefficient vector $y_t \in \mathbb{R}^n$ (next-period returns, $\|y_t\|_\infty \le y_{\max}$), with a latent regime $r_t \in \{1, \ldots, R\}$ evolving in persistent blocks. Regimes recur: the e-th maximal active spell of regime r is *episode* (r, e), a stationary environment with law $P_{r,e}$ over (x, y) pairs; the dormancy $D(r)$ of a regime is the length of the inactive spell between consecutive episodes. Crisis-type regimes have long, irregular dormancy and few historical episodes—the regime that most needs a calibrated specialist is the one observed least and least recently.

Definition 1 (Decision layer and regret)**.** *The feasible set is the cardinality-constrained long-only portfolio $Z = \{z \in [0, w_{\max}]^n : \sum_i z_i \le W,\ \|z\|_0 \le k\}$ with linear utility $F(z, y) = y^\top z$. The plug-in policy decides $z(\hat{y}) = \arg\max_{z\in Z} \hat{y}^\top z$ (exactly solvable: $w_{\max}$ on the k largest positive coordinates when $W \ge k\, w_{\max}$), and the per-step* decision regret *is*

$$\rho(\hat{y}; y) \;=\; F(z^*(y), y) - F(z(\hat{y}), y) \;\in\; [0, B], \qquad B = 2k\, w_{\max}\, y_{\max}.$$

Regret—not prediction error—is the object of interest: a predictor can have large mean-squared error and zero regret (errors that do not change the top-k ranking are free) and small error but large regret (errors concentrated at the selection boundary). This is the core predict-then-optimize lesson (Elmachtoub and Grigas, 2022; Mandi et al., 2024) and the reason the structure diagnostic of Section 5.3 must be re-run under the decision metric rather than inherited from prediction-metric audits.

Specialists are linear decision-focused predictors per regime (*heads*): head $w \in \mathbb{R}^d$ predicts $\hat{y} = X_t w$ and trains on the SPO+ surrogate (Elmachtoub and Grigas, 2022) of Definition 1's regret, whose subgradient requires only two calls to the decision oracle per sample. Episode risk is $L_{r,e}(w) = \mathbb{E}_{P_{r,e}}[\rho(Xw; y)]$.

3.2 Capacity and memory models

Each specialist maintains at most $K < R$ regime heads, under two memory models spanning the discrete and graded readings of capacity (Li, 2026): **hard (LRU)**: learning a non-held regime at full capacity evicts the least-recently-used head (reset to prior), modeling slot-like capacity (an analyst's active playbooks, a finite adapter store); **soft (interference)**: a shared backbone with per-regime adapters in which every foreign update decays untouched adapters' effective knowledge by a factor $(1 - \rho_{\text{int}})$, modeling parameter interference in shared representations. Episode buffers (bounded: the most recent E_{buf} episodes, ≤ 40 days each) count toward the same budget and are evicted with their head: knowledge lost is data lost, not merely parameters reset.

3.3 The RISP mechanism

RISP runs N specialists through a batched competition cycle (Algorithm 1):

1. **Shadow evaluation.** Every specialist produces (shadow) decisions for the current regime r_t; per-specialist decision regret accumulates over a rolling *competition window* of W_c trading days within the regime.
2. **Batched winner-take-all.** At window close, the *window winner* minimizes cumulative window regret plus a niche shaping term $-\lambda(\alpha_{i,r_t} - 1/R)$; only the winner applies one exponentiated-gradient (Hedge) update to its niche affinity $\alpha_i \in \Delta^R$ (Kivinen and Warmuth, 1997; Freund and Schapire, 1997) and trains on the window's data. Batching is the financial adaptation: per-period regret differences are noise-dominated at realistic signal-to-noise, and a W_c-day cumulative comparison drives the wrong-winner probability down exponentially in W_c (Remark 2); $W_c \in \{1, 5, 20, 60\}$ is ablated in Section 5.8.
3. **Invariant niche training.** A specialist updating regime r minimizes, over its episode buffer for r,
$$\mathcal{L}_r(w) \;=\; \mathrm{Var}_e\big[\bar{\ell}_{r,e}(w)\big] \;+\; \beta\,\mathbb{E}_e\big[\bar{\ell}_{r,e}(w)\big], \tag{1}$$
where $\bar{\ell}_{r,e}$ is the mean SPO+ surrogate on stored episode e: the regime-indexed Inv-PnCO objective (Yan et al., 2026; Krueger et al., 2021) with episodes as environments. ERM ablations minimize the mean alone.
4. **Pinning.** When a specialist's affinity for a regime exceeds 0.95, the assignment is *pinned*: the owner alone serves and trains that regime thereafter, making the converged assignment reward-independent by construction (competition re-opens only if the regime inventory changes).

Regime labeling. All arms (ours and baselines) receive the same causal regime label r_t, computed from information through $t-1$ only: **L1** (headline), a rolling-percentile volatility band crossed with trend sign; **L2**, a causal two-state filtered volatility model (forward-only Hamilton-style filter (Hamilton, 1989)) crossed with trend sign. The comparison is therefore information-symmetric (task-incremental in continual-learning terms (Li, 2026)): what differs across arms is solely whether the induced capacity assignment chases current reward, not what each is told.

Episode scarcity. Crisis regimes yield 5–7 historical episodes in long equity samples and fewer in shorter panels; $\mathrm{Var}_e[\cdot]$ over so few draws is fragile. Two mitigations are part of the default design: *sub-episode environments* (non-overlapping blocks within episodes enlarge the environment set while preserving between-episode heterogeneity) and *cross-asset episodes* (the same crisis in different asset classes counts as distinct environments, upgrading the claim to cross-asset invariance within a regime). Our synthetic heterogeneity sweep (Section 5.7) controls the episode count directly; the estimation price of few episodes appears explicitly in Lemma 1.

Algorithm 1 RISP (batched competition cycle)

Require: N specialists with capacity K, regimes $\{1..R\}$, window W_c, EG rate η, niche bonus λ, invariance weight β
Initialize $\alpha_i \leftarrow \mathbf{1}/R$; heads empty; no pins
for $t = 1, 2, \ldots, T$ **do**
  Observe X_t, regime label r_t (causal labeler); serve decision of pinned owner if r_t pinned, else of $\arg\max_i \alpha_{i,r_t}$
  if r_t pinned **then**
    Owner stores (X_t, y_t) in episode buffer; SGD steps on Eq. (1)
  else
    Accumulate window regret $q_i \mathrel{+}= \rho_i$ for all i; buffer the day
    if window length $= W_c$ or regime block ends **then**
      $i^* \leftarrow \arg\max_i \big[-q_i/|\text{win}| + \lambda(\alpha_{i,r_t} - 1/R) \big]$
      EG update: $\alpha_{i^*,r_t} \mathrel{*}= e^{\eta}$; renormalize α_{i^*}
      i^* trains on the window's data (Eq. (1)); LRU/interference bookkeeping
      if $\alpha_{i^*,r_t} \geq 0.95$: pin $r_t \to i^*$
    end if
  end if
end for

4 Theory: A Decomposition of Post-Reactivation Regret

We state the assumptions, the two lemmas that control each axis, and the propositions that assemble them. Proofs not given inline appear in the companion deep-dive document; all results are idealized-model statements in the register of the GAUSE theory (Li, 2026)—guides to mechanism, with the quantitative claims resting on the controlled experiments. We do *not* claim adversarial-regret guarantees for the coupled winner-take-all dynamics.

Assumption 1 (Episode exchangeability; regime-indexed invariance)**.** *For each regime r, episode environments are i.i.d. draws $\xi_{r,1}, \xi_{r,2}, \cdots \sim Q_r$ from a regime-level hyper-distribution, and the deployed (next) episode is a fresh draw. For a fixed head w define $\mu_r(w) = \mathbb{E}_{\xi\sim Q_r}[L_\xi(w)]$ and $\sigma_r^2(w) = \mathrm{Var}_{\xi\sim Q_r}[L_\xi(w)]$.*

This is the regime-indexed form of the Inv-PnCO environment assumption (Yan et al., 2026): "the next crisis is a new draw from the crisis distribution." It idealizes on two counts—real episodes are neither independent (macro memory) nor exchangeable (secular drift)—and we flag both in Section 7.

Assumption 2 (Dormancy and foreign updates)**.** *While regime r is dormant for D steps, an allocation mechanism A routes learning updates among specialists. Under a* reward-driven *A, the specialist holding r is made to learn a currently active regime with per-step probability at least $p > 0$ (active regimes are the only sources of reward, and reward attracts capacity). Under a* reward-independent *A (pinned converged competition, fixed niches, frozen gate, hand-designed silos), the assignment is measurable with respect to information independent of the realized reward stream, and the owner of r receives no foreign updates during r's dormancy.*

4.1 The invariance axis

Lemma 1 (Episode transfer)**.** *Under Assumption 1, for any fixed w and $\delta \in (0, 1]$, with probability $\geq 1 - \delta$ over the test episode $e' \sim Q_r$,*

$$L_{r,e'}(w) \;\leq\; \mu_r(w) + \sigma_r(w)\sqrt{(1-\delta)/\delta} \;\leq\; \mu_r(w) + \sigma_r(w)/\sqrt{\delta}.$$

Moreover, with E_r training episodes and $L \in [0, B]$, the empirical mean satisfies $|\hat{\mu}_r(w) - \mu_r(w)| \leq B\sqrt{\log(2/\delta)/(2E_r)}$ with probability $\geq 1 - \delta$ (Hoeffding over episodes), and an analogous $O(B/\sqrt{E_r})$ bound holds for the empirical standard deviation via empirical-Bernstein concentration (Maurer and Pontil, 2009).

Proof. The first display is Cantelli's one-sided inequality applied to the random variable $L_\xi(w)$, $\xi \sim Q_r$: $\Pr[L_\xi \geq \mu_r + \tau] \leq \sigma_r^2/(\sigma_r^2 + \tau^2)$, solved at the δ level. The second is Hoeffding's inequality for $[0, B]$-bounded i.i.d. variables. □

The RISP objective (Eq. 1) is, up to the $O(B/\sqrt{E_r})$ estimation terms, a Lagrangian scalarization of the bound's two arguments: sweeping β traces the (μ_r, σ_r) Pareto frontier, and for each tail level δ some $\beta(\delta)$ optimizes the bound. ERM ignores σ_r entirely and can sit far up the variance axis—the mechanism by which it loads on episode-specific regularities (Section 5.7 constructs this failure explicitly). We are explicit about register: the trained $\hat{w}$ depends on the training episodes, so applying Lemma 1 to $\hat{w}$ is a plug-in heuristic at the same level as Inv-PnCO's use of its Theorem 1, not a uniform-convergence guarantee; the estimation terms are the formal price of few episodes.

Lemma 2 (Divergence-to-regret bridge)**.** *For any episode laws P, P' on $\mathcal{X} \times \mathcal{Y}$ and any measurable head w,*

$$|L_P(w) - L_{P'}(w)| \;\leq\; B \cdot \mathrm{TV}(P, P') \;\leq\; B\sqrt{\mathrm{KL}(P \,\|\, P')/2}.$$

Consequently, if a representation renders episode laws ε-invariant in KL (as in Inv-PnCO-style guarantees, which bound divergences between solution distributions (Yan et al., 2026)), then $\sigma_r(w) \leq B\sqrt{\varepsilon/2}$ for heads factoring through it, and Lemma 1 applies with that variance.

Proof. Regret is $[0, B]$-valued pointwise (Definition 1), so the difference of its expectations under P, P' is at most $B \cdot \mathrm{TV}$ by the variational characterization of total variation applied to $g = \rho/B \in [0, 1]$; Pinsker's inequality (Tsybakov, 2009) gives $\mathrm{TV} \leq \sqrt{\mathrm{KL}/2}$. □

This closes the announced technical gap: invariance statements in the divergence metric transfer to the regret metric at the explicit price of the regret range $B = 2k\, w_{\max}\, y_{\max}$ (loose—empirical regret ranges are roughly fifty times smaller—but explicit; the decomposition statement below survives verbatim in the KL metric if the constant is objectionable).

4.2 The retention axis

Lemma 3 (Eviction rate under reward-driven allocation)**.** *Hard memory model, capacity K, Assumption 2 with $W_a \geq K$ distinct active regimes during the dormancy of r. Let U_D be the number of distinct non-held foreign regimes the owner of r is made to learn during D dormant steps. Then r's head is evicted on the event $\{U_D \geq K\}$, and $\Pr[U_D < K] \leq \Pr\left[\mathrm{Binom}(D,\, p\,(W_a - K + 1)/W_a) < K\right] \to 0$ exponentially in D. Under the soft model, the retained strength of r after N_D foreign updates is $(1 - \rho_{\mathrm{int}})^{N_D} \to 0$.*

Proof. Each dormant step independently delivers a foreign-regime update with probability $\geq p$; conditional on an update, a not-yet-held distinct regime is hit with probability $\geq (W_a - K + 1)/W_a$ while fewer than K distinct regimes have been accumulated. The waiting time to K distinct hits is thus stochastically dominated by a sum of K geometric variables, giving the binomial tail; LRU evicts r once K distinct foreign heads displace it. The soft case is immediate from once-per-update multiplicative decay. □

Proposition 1 (Reward-independence zeroes the forgetting deficit)**.** *Define the forgetting deficit $\Phi(D; A) = \Pr[r\text{'s head lost at reactivation}]$ (hard) or $1 - \mathbb{E}[(1 - \rho_{\mathrm{int}})^{N_D}]$ (soft). Under Assumption 2: (i) for any reward-driven A, $\Phi(D; A) \to 1$ as $D \to \infty$ at the rate of Lemma 3; (ii) for any reward-independent covering A, $\Phi(D; A) = 0$ for all D: the owner of r applies zero foreign updates during dormancy, so its head—and its episode buffer—are exactly preserved (hard) or decay only with the regime's own drift (soft).*

Proof. (i) is Lemma 3. (ii): by reward-independence the assignment of r does not respond to the active stream; by coverage the owner exists and holds r within capacity; the owner is selected to learn only its own niche set, none of which is active foreign load, so $U_D = 0$ deterministically and no interference decay is triggered. □

The two regimes are different *functions* of D—one tends to 1, the other is identically 0—which is what makes the dichotomy experimentally sharp (Section 5.6). Coverage is the second conjunct, inherited from GAUSE: random fixed niches are reward-independent but can leave a regime unowned, retaining only what they cover; emergent competition guarantees coverage at $N \geq R$ (Li, 2026), which we re-verify in our setting (every seed converges to a one-per-regime covering assignment; Section 5.2).

4.3 Assembly

Proposition 2 (Post-reactivation regret decomposition)**.** *Under Assumptions 1–2, let regime r reactivate after dormancy D into a fresh episode e', served by allocation A with the owner's head $\hat{w}_r$ trained on E_r episodes. Let C_{relearn} denote the probe-window regret excess of a from-prior head over a converged retained head (an empirical constant of the learning problem). Then, at tail level δ,*

$$\mathbb{E}[\rho_{\mathrm{react}}(r)] \;\le\; \underbrace{\mu_r(\hat{w}_r) + \frac{\sigma_r(\hat{w}_r)}{\sqrt{\delta}} + O\Big(\frac{B}{\sqrt{E_r}}\Big)}_{G_{\mathrm{inv}}} \;+\; \underbrace{\Phi(D;A)\,C_{\mathrm{relearn}}}_{G_{\mathrm{forget}}}.$$

Proof. Condition on the retention event. On retention (probability $1-\Phi$), the served head is $\hat{w}_r$ and Lemma 1 bounds its regret on the fresh episode; on eviction (probability Φ), the served head is fresh and its probe-window excess over the retained benchmark is C_{relearn} by definition. Combine, then upper-bound the retention branch's coefficient $(1-\Phi)$ by 1. □

Proposition 3 (Independent controls; the 2×2 prediction)**.** *G_{forget} depends only on $(A, D, K, \textit{memory model})$ and is zeroed by reward-independent covering assignment regardless of the training objective; G_{inv} depends only on $(\textit{objective}, Q_r, E_r)$ and is optimized by the variance-penalized objective regardless of the assignment. The bound is therefore additive in the two design axes, predicting monotone improvement along both axes of the 2×2. Moreover the realized interaction is predicted* super-additive *in one specific cell: under a reward-driven A with $\Phi \approx 1$, the served head at reactivation is fresh regardless of its training objective, so the invariance axis is inert in the reward-driven row (G_{inv} improvements act on a head that is not there to serve).*

Remark 1 (Honest register)**.** *Additivity of the* bound *does not force additivity of realized regret; we therefore measure the empirical interaction index $I = (A1-A5)+(A1-A7)-(A1-A6)$ and report it with the headline (Section 5.2). The super-additivity prediction—invariance inert in the reward-driven row—is the distinctive falsifiable signature separating our account from a generic "both tricks help" story.*

Proposition 4 (Break-even dormancy)**.** *Charge a carrying cost c per idle specialist-day (infrastructure, attention, calibration operations) and let V scale one unit of probe-window regret to portfolio value over the N_p-day probe. Retention of regime r's specialist pays over a dormancy-reactivation cycle iff*

$$c \cdot D \;\le\; \Phi(D; A_{\mathrm{rd}})\, C_{\mathrm{relearn}}\, N_p\, V,$$

with Φ the reward-driven deficit of Lemma 3. Since Φ saturates at 1 while cD grows linearly, the paying region is a window *$[D_{\min}, D_{\max}]$: dormancy long enough that eviction is near-certain, short enough that cumulative carrying cost does not swamp the one-shot reactivation saving. Both endpoints are computed from measured quantities in Section 5.6.*

Remark 2 (Batching the competition: a prediction revised by experiment)**.** *The window winner is decided by a W_c-day cumulative regret comparison. For two specialists with true within-regime quality gap Δ and per-day comparison noise σ_n (sub-Gaussian), Hoeffding gives* $\Pr[\textit{wrong winner in one window}] \le \exp(-W_c\Delta^2/2\sigma_n^2)$, *which motivated our pre-registered expectation that per-step competition ($W_c = 1$) would converge to noise at realistic daily signal-to-noise. The experiment revises this: the assignment is decided not by any single window but by the* accumulation *of EG updates across many windows, and that accumulation performs the averaging the single-window analysis assigns to batching—post-reactivation regret is flat across $W_c \in \{1, 5, 20, 60\}$, with $W_c = 1$ marginally best (Section 5.8). The single-window bound is correct but answers the wrong question; the operative statistic is the win-rate differential integrated by the affinity dynamics. We retain $W_c = 20$ as the default for interpretability (one affinity update per fortnight is auditable) and report the sweep.*

5 Experiments

5.1 Protocol, arms, and metrics

Design philosophy. Every claim is pre-registered as an ordering prediction before the run, tested across 20 seeds with Welch tests and Holm–Bonferroni correction, and reported whether or not it survives—including two predictions that did *not* (the batching expectation, Remark 2, and the real-data dissociation, Section 5.3). All numbers trace to released per-experiment result files. We use three data tiers, in decreasing order of experimental control and increasing order of realism: (i) *synthetic* regime-switching markets with known ground truth (the headline, because only here is every causal claim testable); (ii) *semi-synthetic* regime-stitched real data (real per-regime cross-sections re-arranged into controlled dormancy schedules); (iii) *real walk-forward* panels (five Bybit USDT crypto pairs, daily, 2021–2025; FRED commodities, 2015–2025). We state plainly that the roadmap's equities flagship (CRSP S&P 500 constituents, 1990–2025, with 5–7 crisis episodes) is a planned extension requiring data access not available at the time of writing; nothing here is presented as equity evidence.

Synthetic market. $n = 20$ assets, $R = 4$ regimes (three common, one rare "crisis"), $d = 20$ features per asset: two *invariant* dimensions carrying the regime's decision-relevant structure θ_r (distinct, well-separated across regimes, stable across episodes), one *spurious* dimension whose loading $\gamma_{r,e}$ is redrawn each episode (the episode "weather"), sixteen distractors, and a constant. Signal-to-noise is calibrated to realistic daily cross-sections ($|\theta_r| \approx 0.012$ against noise 0.02–0.06; crisis noise $3\times$ calm), which makes fresh relearning genuinely slow—the regime in which the allocation question is non-trivial, and, as our audit shows (Section 5.9), the regime in which it *stays* non-trivial. Common regimes alternate in blocks of 25–60 days; the crisis regime appears for 15–30 days roughly every fourth cycle, giving dormancy spells of 250–450 days and 15 evaluated reactivations per run ($T \approx 2{,}370$; first half burn-in, second half evaluated). Decision layer: $k = 4$ of 20, $w_{\max} = 0.25$.

Arms. All learner arms share the identical specialist class (linear SPO+ heads, capacity $K = 2$ of $R = 4$ unless swept, identical learning rates and buffers); they differ *only* in allocation and objective. A1 capacity-K monolith (ERM; always trains the active regime—the canonical reward-driven allocation and the regime-conditioned finance baseline under a capacity budget). A2 mixture-of-experts with a gate trained on realized regret (reward-driven routing). A3 recent-performance capital allocator: capital shares softmax-weight trailing 60-day performance; each specialist trains on the active regime with probability equal to its capital share (capital is recalibration intensity—the industry baseline; its dormant-specialist erosion rate is governed by the softmax floor). A4 random fixed niches (reward-independent, coverage gaps possible). A5 RISP-ERM (emergent assignment, ERM training). A6 RISP-full (emergent assignment, invariant objective, Eq. 1). A7 monolith with the invariant objective (tests whether the objective can substitute for allocation). A8a Hedge over fixed strategies; A8b Hedge over capacity-bounded learning experts (training follows the top weight)—the adaptive online-learning pair of the scope statement. A9 oracle-pinned assignment with ERM; A10 oracle-pinned with the invariant objective (the skyline: hand-given perfect assignment, nothing to discover).

Metrics. *Post-reactivation regret*: mean daily decision regret over the first 15 days of every block whose regime was dormant ≥ 90 days (≥ 80 in the stitched-crypto schedules, whose blocks are shorter; the probe window mirrors the GAUSE protocol). *Overall*: mean daily regret over the evaluation half. *Steady*: within-block regret beyond the probe window (the converged-service level; its floor across arms estimates irreducible noise). Where percentages of "excess" regret are quoted, the floor is the steady-state regret of the A10 skyline (0.0250)—the best converged service achievable in this market—and we always quote raw numbers alongside.

5.2 E1 — the headline 2×2 dissociation

Table 1 and Figures 1–2 give the headline. Every pre-registered prediction is confirmed (Holm-corrected Welch tests over 20 seeds):

The retention axis is real. Emergent reward-independent assignment with identical ERM training cuts post-reactivation regret from 0.0342 (A1) to 0.0321 (A5), -6.1%, $p = 1.3 \times 10^{-3}$ (Holm 1.0×10^{-2}). Random fixed niches (A4) sit statistically with A5 ($p = 0.28$) but with twice the seed variance—the signature of coverage gaps: a seed whose random draw covers all four regimes retains like A5; a seed with an unowned regime pays for it on every reactivation.

The invariance axis is real—inside a retaining allocation. Holding the emergent assignment fixed, the episode-variance objective cuts post-reactivation regret from 0.0321 (A5) to 0.0307 (A6), -4.4%,

Table 1: E1: post-reactivation, overall, and steady-state decision regret (mean ± 95% CI over 20 seeds; synthetic market, K=2, hard memory). Horizontal rules group: skylines / RISP / retaining baselines / reward-driven arms / adaptive pair. Pre-registered ordering confirmed: A6 $<$ A5 $\approx$ A4 $<$ A1 $\approx$ A2 $\approx$ A7 $<$ A3, with A8a worst overall and A8b tracking the monolith.

Arm	**Post-react.**	**Overall**	**Steady**
A10 oracle assignment + INV (skyline)	0.0305 ± 0.0008	0.0259 ± 0.0006	0.0250
A9 oracle assignment + ERM	0.0321 ± 0.0008	0.0274 ± 0.0007	0.0262
A6 RISP-full (emergent + INV)	$\mathbf{0.0307 \pm 0.0007}$	0.0260 ± 0.0006	0.0251
A5 RISP-ERM (emergent + ERM)	0.0321 ± 0.0007	0.0275 ± 0.0006	0.0264
A4 random fixed niches (ERM)	0.0330 ± 0.0013	0.0283 ± 0.0011	0.0272
A1 monolith (ERM)	0.0342 ± 0.0010	0.0269 ± 0.0004	0.0246
A7 monolith (INV)	0.0344 ± 0.0010	0.0266 ± 0.0004	0.0243
A2 MoE router (regret-trained gate)	0.0341 ± 0.0011	0.0270 ± 0.0005	0.0248
A3 recent-performance allocator	0.0378 ± 0.0010	0.0294 ± 0.0006	0.0269
A8a Hedge over fixed strategies	0.0439 ± 0.0009	0.0382 ± 0.0007	0.0370
A8b Hedge over learning experts	0.0341 ± 0.0010	0.0267 ± 0.0005	0.0244

$p = 6.1 \times 10^{-3}$ (Holm 4.3×10^{-2}). Jointly, A6 improves on A1 by 10.3% raw ($p = 9.2 \times 10^{-7}$), which is 38% of the excess over the irreducible noise floor ($0.0342 - 0.0250 = 0.0092$ down to 0.0057).

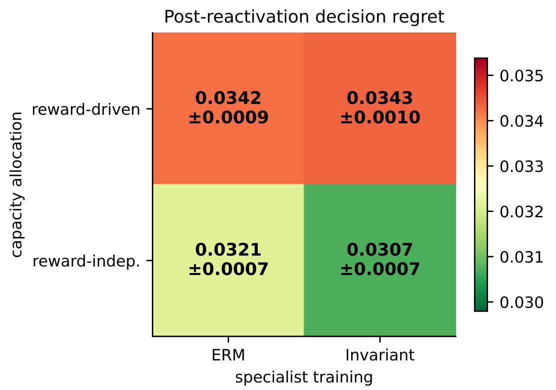
The interaction is super-additive, exactly as Proposition 3 predicts. The invariant objective in a reward-driven monolith (A7) is indistinguishable from ERM (A7 vs. A1 post-reactivation: $p = 0.88$): an objective cannot help a head that was evicted during dormancy. Per-seed, the interaction index $I = (\text{A1}{-}\text{A5}) + (\text{A1}{-}\text{A7}) - (\text{A1}{-}\text{A6}) = -0.0015 \pm 0.0005$ (95% CI excluding zero): the joint improvement exceeds the sum of the marginal improvements because the invariance margin only exists where retention preserves the head it trained. This asymmetry is the distinctive signature of the two-mechanism account—a generic "both tricks help additively" story would put I at zero, and a shared-cause story would make A7 help. (A7 is directionally better than A1 *overall*—0.0266 vs. 0.0269, not significant—consistent with the objective helping while heads are alive and being erased at each eviction.)

Reward-driven arms cluster together, as the theory says they must. The learned router tracks the monolith ($p = 0.83$); Hedge over learning experts tracks it too (0.0341); the recent-performance allocator is *worst* among learner arms (0.0378)—it not only forgets dormant regimes but trains specialists on a capital-weighted blend of foreign regimes throughout, paying interference even in steady state. The sleeping-experts reading of our problem is answered constructively: A8a (fixed strategies) retains trivially yet is dominated by every learning arm everywhere (0.0439 post-reactivation, 0.0370 steady)— retention without within-niche learning is worth little; A8b (learning experts) learns but forgets—adaptivity of the *combination* does not protect the *experts*. The contribution lives exactly in the gap this pair brackets.

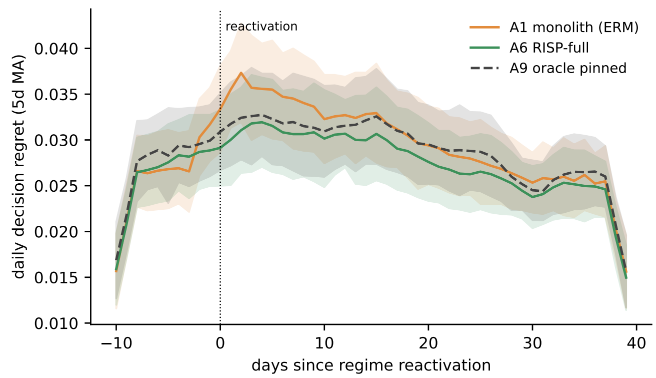
RISP matches the skyline it was never given. A6 is statistically indistinguishable from A10—the hand-pinned oracle assignment with the same invariant objective (0.0307 vs. 0.0305, $p = 0.72$)—and nominally better than the oracle assignment with ERM training (A9, $p = 0.011$, marginal after Holm at 0.057). As in GAUSE, the value of competition is not beating hand-assignment given the labels (it cannot); it is *recovering the oracle partition automatically*: in every seed the population converges to a covering one-per-regime assignment (the three common regimes pin at affinity ≥ 0.95; the rare crisis regime reaches 0.67–0.87—short of the pin threshold because crisis windows are scarce—but with the correct unique owner serving throughout, an honest wrinkle we report rather than hide).

5.3 E0 — the structure diagnostic on real data, and an honest null

Before any regime-aware mechanism is credited on real data, the diagnostic-first protocol requires evidence that regime structure *exists on the decision metric*: an oracle regime-conditioned predictor (one ridge model per causal regime label, walk-forward refit) must beat the same machinery under block-shuffled labels. GAUSE's oracle audit found crypto and commodities carry no exploitable regime structure under the *prediction* metric



(a) The 2×2: the invariance axis is inert in the reward-driven row (the predicted asymmetry).



(b) Daily regret aligned on reactivation.

Figure 1: **Left:** post-reactivation decision regret across allocation × objective (mean ± 95% CI, 20 seeds). Cells: A1 / A7 (top), A5 / A6 (bottom). **Right:** mean daily regret around reactivations (5-day MA, 95% CI): the monolith pays a relearning transient concentrated in the probe window; RISP-full tracks the oracle-assignment skyline from day one.

Table 2: E0: structure diagnostic on the decision-regret metric (walk-forward, second-half evaluation; shuffled = block-label permutation, 10 draws). Negative gap = conditioning is (insignificantly) *worse* than shuffled. No domain–labeler pair passes; the real-data dissociation is therefore predicted null, and is (Section 5.4).

Domain	**Labeler**	**Real**	**Shuffled**	**Gap**	z
Crypto (5 pairs, daily)	L1 vol×trend	0.01220	0.01202	−1.5%	−0.66
Crypto (5 pairs, daily)	L2 filtered HMM	0.01212	0.01204	−0.7%	−0.17
Commodities (3, daily)	L1 vol×trend	0.00834	0.00832	−0.3%	−0.14
Commodities (3, daily)	L2 filtered HMM	0.00836	0.00838	+0.2%	+0.16

(Li, 2026); since prediction-optimal and decision-optimal differ (the core predict-then-optimize lesson), we pre-registered the re-test under decision regret as a finding either way.

The answer (Table 2) is unambiguous: **no exploitable regime structure on the decision metric** in either domain, under either causal labeler—regime-conditioned regret is statistically identical to block-shuffled-label regret (all $|z| < 0.7$), and both are identical to a pooled unconditioned model. The prediction-metric null of GAUSE *extends* to the decision metric here: the PnO distinction, which in principle could resurrect structure invisible to prediction error, does not do so on these panels. We bound the claim honestly: these are small universes (five crypto pairs, three commodities) at daily frequency with a standard momentum/volatility feature inventory and two labelers; richer features, intraday horizons, or larger cross-sections could yet pass the gate. What the protocol forbids is claiming the mechanism's benefits where the gate fails—which is exactly what a less guarded evaluation would have produced, since the machinery happily trains and "specializes" on structureless labels (the GAUSE negative control: organization is not structure).

5.4 E1s — the real-data dissociation is null, as the failed gate predicts

We ran the full eleven-arm comparison on *regime-stitched* real crypto data: contiguous same-label day-blocks of the actual cross-sections (real returns, real features), re-arranged into controlled schedules with the rare regime dormant ≈ 200 days (20 seeds). All learner arms are statistically indistinguishable (post-reactivation regret 0.0134–0.0141; every pairwise $p > 0.2$). Under a failed E0 gate this is the *predicted* outcome—there is no per-regime structure for a specialist to retain, so retention cannot pay—and we present it as the protocol working, not as the mechanism failing: the same arms, the same code, and the same statistics that produce the sharp dissociation of Table 1 produce a flat table when the precondition is absent. A method whose

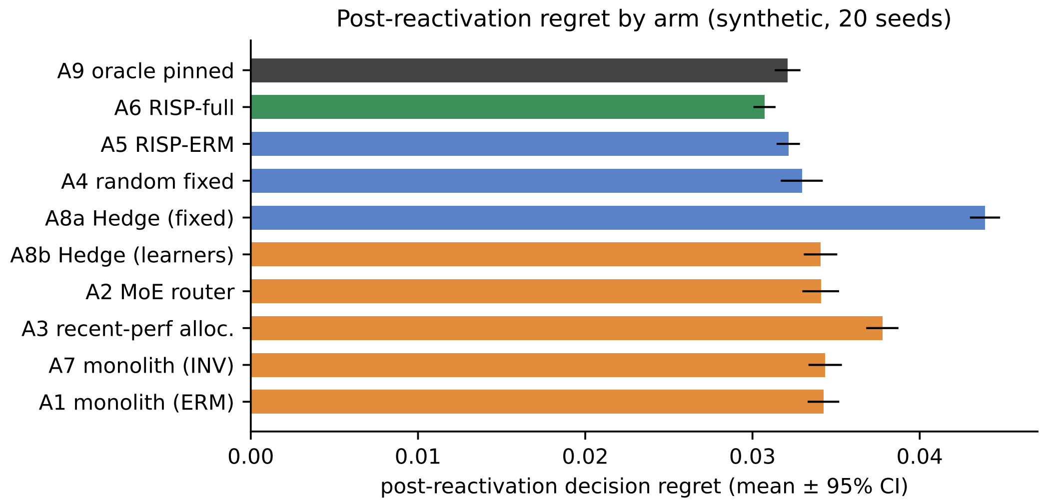


Figure 2: E1, all eleven arms (synthetic, K=2, hard memory, 20 seeds). Color encodes class: green = emergent reward-independent + invariant; blue = reward-independent (ERM or fixed); orange/red = reward-driven; black = oracle skyline. The dissociation separates classes, not architectures: every reward-chasing arm clusters with the monolith regardless of its machinery.

evaluation cannot go flat on structureless data should not be trusted when it goes sharp. (Fallback F1 of the pre-registered design: where the gate fails on all available real domains, the headline is the controlled study plus the diagnostic protocol; the equities panel, with its richer episode inventory, is the planned real-data test.)

5.5 E2 — capacity sweep: when does the dissociation close?

Sweeping per-specialist capacity $K = 1, \ldots, 4$ $(= R)$ under both memory models (Figure 3): RISP arms are *flat in* K (A6: 0.0307 at every K; A5: 0.0321)—a converged one-per-regime assignment needs one slot per specialist, and extra capacity is simply unused. Under the hard model the reward-driven arms improve monotonically with K and approach the retaining class only at $K = R$ (monolith: $0.0343 \to 0.0322$, matching A5 at K=4; router likewise), where eviction becomes unnecessary and the allocation question dissolves. The soft (interference) model inverts the diagnosis instructively: with no eviction, capacity slots stop binding and the reward-driven arms are *flat* in K at ≈ 0.0328—gentler than hard eviction at tight capacity (0.0343 at $K \leq 2$) but never closing the gap to A6 (0.0307) at *any* capacity, because graded interference from foreign updates erodes dormant heads no matter how many slots exist. Slack capacity is thus one audited boundary of the contribution, with a caveat that depends on the memory physics: with $K \geq R$ *and* hard isolation, conditioning alone suffices (the regime-switching-finance baseline is vindicated exactly there); with shared representations (the realistic reading for parametric models), interference keeps the population's structural protection valuable at every capacity.

5.6 E3 — dormancy sweep and the break-even window

Controlled schedules place the focal regime's reactivations after dormancy $D \in \{21, 63, 126, 252, 504\}$ days (Figure 4a). The retention gap (A1 − A6 post-reactivation) grows from 0.0015 at $D = 21$ to 0.0038–0.0044 for $D \geq 63$ and saturates—the signature of Lemma 3: once dormancy exceeds the eviction scale ($\approx K$ foreign-regime acquisitions, here a few dozen days), the monolith's head is gone with probability ≈ 1 and longer dormancy adds no further loss, while the RISP owner's deficit is flat at zero by construction. The recent-performance allocator degrades fastest (its capital floor keeps eroding the dormant specialist throughout). Feeding the measured quantities into Proposition 4 (relearning cost $C_{\text{relearn}} \approx 0.004$ per probe day over $N_p = 15$ days): at a carrying cost of 1% of the per-day probe saving per idle-specialist-day, retention

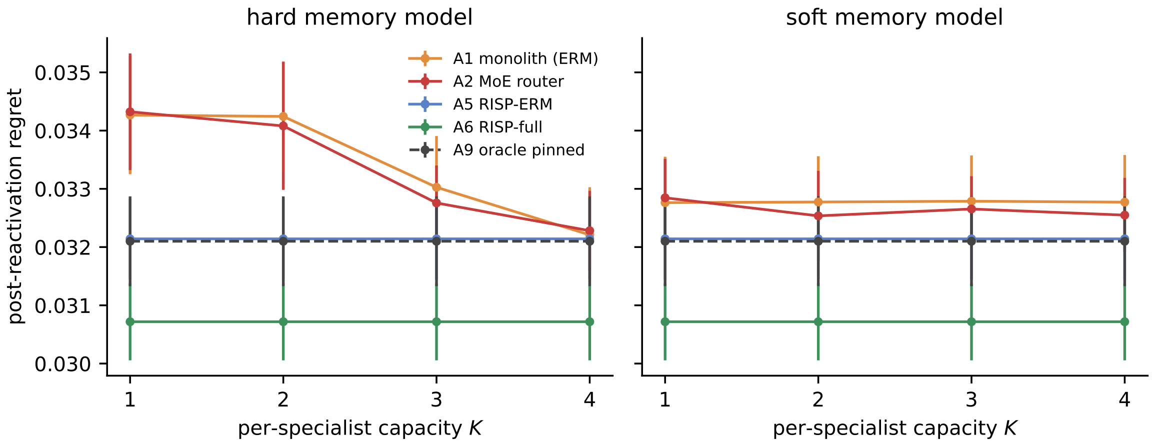


Figure 3: E2: post-reactivation regret vs. per-specialist capacity K ($R = 4$; 20 seeds; left: hard LRU, right: soft interference). Reward-independent arms are flat in K; reward-driven arms close the gap only as $K \to R$ under hard slots, and never fully close it under interference.

pays for D between roughly 40 and 6,000 days; at 10% carrying cost the window shrinks to $D \in [\approx 60, \approx 600]$ days—and at carrying costs above $\approx 15\%$ of the probe saving, *no* dormancy length pays. The existence of the *upper* break-even is the practically honest point: idle desks are not free, and a fund whose crisis playbook reactivates once a decade may rationally let it decay if maintaining it costs enough.

5.7 E4 — episode heterogeneity isolates the invariance axis

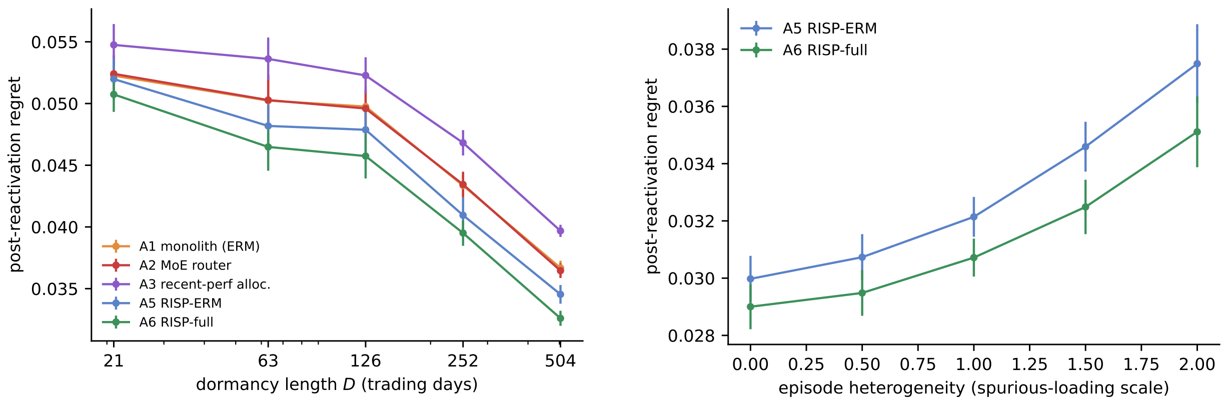
Holding the allocation fixed (emergent, retaining) and sweeping the episode-heterogeneity scale (the spurious loading's standard deviation relative to invariant signal) from 0 to 2× baseline (Figure 4b): the ERM specialist degrades steadily ($0.0300 \to 0.0375$ post-reactivation) while the invariant specialist degrades about 23% slower ($0.0290 \to 0.0351$), so the A5−A6 gap widens monotonically from 0.0010 to 0.0024 (2.4×)—the regime-indexed analogue of the Inv-PnCO heterogeneity result, and direct evidence that the invariance margin is purchased against episode variability specifically (at zero heterogeneity the objectives nearly coincide; the small residual gap is the variance penalty acting as a regularizer against the sixteen distractor features). Reactivation is an out-of-distribution event *within* one's own regime to precisely the degree that episodes differ; retention solves dormancy, and this axis is what remains after.

5.8 E5 — ablations

β sweep, and the variance-only control fails as predicted. Post-reactivation regret is U-shaped in the mean-loss weight: $\beta = 0$ (variance only) gives 0.0356—*worse than retained ERM* (0.0321)—confirming in the decision setting the Inv-PnCO finding that the variance term alone is degenerate (a head can achieve zero episode-variance by being uniformly bad); the plateau $\beta \in [0.25, 4]$ gives 0.0307–0.0317; very large $\beta = 16$ degrades to 0.0352—worse than the retained-ERM arm itself (0.0321), which is the honest reading: β multiplies the effective gradient magnitude, so $\beta = 16$ is not "clean ERM recovered" but an unstably large step on the mean term (A5 remains the correct $\beta \to \infty$ reference, trained at the proper rate). The objective needs both terms, and is insensitive to β across an order of magnitude.

Competition-window sweep. $W_c \in \{1, 5, 20, 60\}$ days yields 0.0303, 0.0306, 0.0307, 0.0309: flat, with per-step competition marginally *best*—the revision of our pre-registered batching expectation discussed in Remark 2. The mechanism's robustness to its main scheduling hyperparameter is itself a deployment asset.

Niche bonus and pinning. Competition alone ($\lambda = 0$) reaches 0.0314 versus 0.0307 with the default shaping ($\lambda = 0.3$)—the bonus accelerates but does not cause the assignment, consistent with GAUSE's λ ablation. Disabling pinning costs 0.0320 versus 0.0307: keeping competition perpetually open re-admits



(a) Dormancy sweep (E3).

(b) Episode-heterogeneity sweep (E4).

Figure 4: **(a)** Post-reactivation regret vs. dormancy D of the focal regime: the reward-driven arms' deficit saturates once D exceeds the eviction scale (Lemma 3); levels are not comparable across D (schedules differ), arm gaps within each D are. **(b)** Fixing the retaining allocation and sweeping episode heterogeneity isolates the invariance axis: the ERM−INV gap grows 2.4× across the sweep.

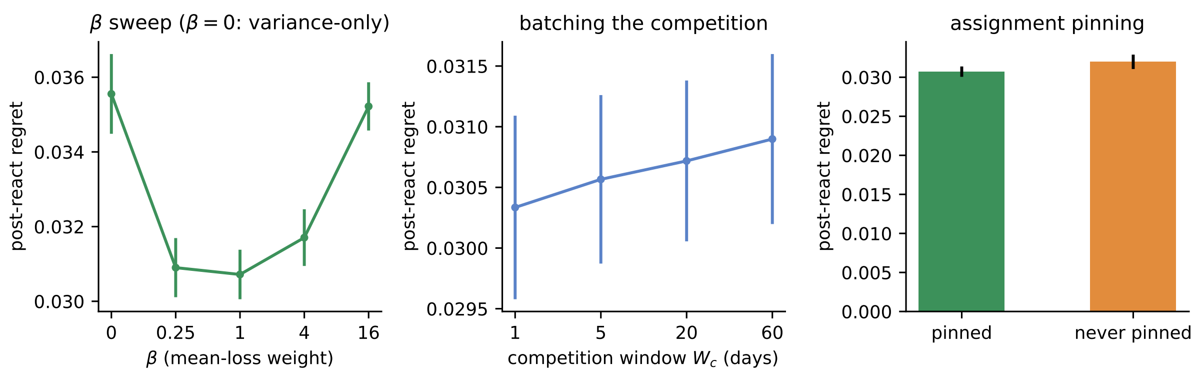


Figure 5: E5 ablations (post-reactivation regret, 20 seeds). **Left:** the invariance objective is U-shaped in β and the variance-only control ($\beta = 0$) fails below retained ERM. **Center:** robustness to the competition window W_c (the revised batching prediction, Remark 2). **Right:** pinning the converged assignment matters.

exactly the reward-coupling the mechanism exists to remove, though the EG dynamics' inertia limits the damage.

5.9 E6 — audit: when retention is worthless

This experiment exists because our first implementation refuted us. At the default parameters of an early prototype (signal-to-noise an order of magnitude above realistic daily cross-sections), *no arm separated from any other*: the monolith relearned an evicted regime within two to five days, the probe window scored mostly converged behavior, and retention was demonstrably worthless. Rather than discard the parameter regime, we promoted it to an audit axis. Sweeping the signal-to-noise multiplier over $\{0.5, 1, 2, 4, 8\}$ (Figure 6):

- The monolith's relearning penalty (post-reactivation over steady regret) grows from 1.28× to 2.06× with SNR—relative relearning cost rises—but its *absolute* steady level collapses ($0.0296 \to 0.0159$), because strong signal is refit almost instantly and then exploited deeply.
- The RISP advantage over the monolith is +8.3% at 0.5×, +10.3% at baseline, +7.0% at 2×—then *inverts*: −13.1% at 4× and −49.3% at 8×. At high SNR the episode-specific loading is itself strongly

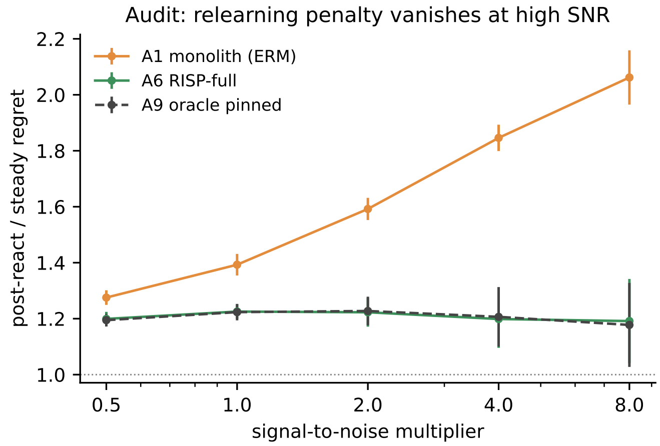


Figure 6: E6 audit: the relearning penalty (post-reactivation regret relative to own steady state) as the signal-to-noise multiplier sweeps 0.5×–8×. The monolith's relative penalty grows, but its absolute steady level collapses; the RISP advantage inverts near 2–3× (see text).

exploitable, and an ERM monolith that greedily refits each episode beats an invariant specialist that on principle refuses to chase it. The skyline arm inverts identically (A9 tracks A6), confirming this is a property of the *regime*, not of our assignment mechanism.

The audit therefore draws the mechanism's boundary as a measured crossover, near 2–3× our baseline calibration: RISP pays where signals are weak enough that relearning is slow and episode-chasing does not compensate—which is the empirically realistic regime for daily cross-sectional return prediction—and is counterproductive where within-episode signal is strong, fast to fit, and worth chasing. This is the decision-domain analogue of GAUSE's stale-specialist result under intra-regime drift: reward-independent retention is an asset under dormancy and a liability under exploitable drift, and the honest deliverable is the location of the boundary.

5.10 Statistical protocol and what we deliberately do not claim

All headline comparisons are Welch tests over 20 independent seeds with Holm–Bonferroni correction within each experiment's pre-registered pair family; tables report mean ± 95% CI. The synthetic and stitched experiments expose every arm to identical data streams per seed (paired design). We do *not* report Sharpe ratios, deflated or otherwise, for the synthetic headline—regret against a known optimum is strictly more informative when ground truth exists—and we make no claim of live trading profitability anywhere: by the E0 gate, the panels where we could have manufactured such claims do not carry the structure that would make them meaningful. For any future real-data deployment passing the gate, the protocol of record is: deflated Sharpe (Bailey and López de Prado, 2014) for any P&L claim, an SPA/Reality-Check test (Hansen, 2005; White, 2000) across the arm family, transaction costs at 5/10/25 bps with borrow, nested walk-forward for every hyperparameter, and the break-even-dormancy carrying-cost analysis of Proposition 4—pre-committed here.

5.11 Synthesis

Across E0–E6 the two-axis account survives every test we could construct against it, and fails exactly where it claims it should fail. Retention is an allocation property: every reward-independent covering arm retains (A4–A6, A9, A10), every reward-chasing arm relearns (A1–A3, A8b), the boundary is crisp ($p \sim 10^{-6}$ for the class separation), and no training objective crosses it (A7). Invariance is an objective property: it pays only inside a retaining allocation (super-additive interaction, CI excluding zero), grows with episode heterogeneity (E4), and requires both of its terms (β sweep). The mechanism needs its preconditions: bounded capacity

(E2), real dormancy (E3), weak-signal relearning (E6), and verifiable regime structure (E0/E1s)—each audited with the experiment that locates its boundary. Where all preconditions held, the emergent system matched the hand-designed skyline without being given the assignment; where any failed, the system honestly bought nothing.

6 Discussion

Why two mechanisms, not one. The temptation in both literatures is monism. Continual-learning work treats non-stationarity as forgetting and reaches for retention; OOD-generalization work treats it as shift and reaches for invariance. The financial setting forces the synthesis because its non-stationarity is genuinely two-layered: regimes *recur* (so there is something worth retaining) and recurrences *differ* (so retention alone retains the wrong thing). The super-additive interaction is the empirical signature that the layers are real: each mechanism's value is contingent on the other having done its job. We conjecture this structure—retain the niche, generalize within it—is generic to any domain with recurring-but-drifting modes: climate regimes, epidemic waves, demand seasons, adversary playbooks.

The allocator, not the org chart. Multi-strategy funds already silo desks by regime—the practitioner instantiation of reward-independent assignment. RISP's contribution to that practice is not the silo but the analysis of what destroys it: recent-performance capital allocation *across* silos (our A3, the industry's default) erodes dormant desks at a rate set by its capital floor, and a learned router—the modern ML upgrade—fails identically and for the same structural reason (Proposition 1). The emergent mechanism additionally shows the silo map need not be designed: at $N \gtrsim R$ it is recovered automatically, matching the hand-pinned skyline, with graceful degradation documented when competition's preconditions weaken (the unpinned crisis niche of Section 5.2).

What the diagnostic-first protocol buys. The most consequential result in this paper for applied work is arguably the null: two real asset classes, two causal labelers, and a decision-metric oracle test that refuses to certify regime structure—after which the full eleven-arm machine, run anyway, produces exactly the flat table the refusal predicts. The protocol converts "our regime-aware method beat baselines on a backtest" (unfalsifiable in the presence of label noise and selection) into two separable, individually falsifiable claims: *structure exists here* (E0, cheap, runs before any method) and *given structure, the mechanism captures it* (E1, controlled). We commend the split to the regime-aware finance literature at large.

7 Limitations

1. **The headline is synthetic; the real-data result is a null.** Our dissociation is demonstrated where its preconditions verifiably hold—controlled markets with known regime structure, episode heterogeneity, and realistic SNR—because on the real panels available to us the E0 gate fails and the dissociation is (correctly) absent. We regard gate-then-claim as the right epistemics, but the constructive real-data demonstration remains undone: the planned equities panel (S&P 500 constituents, 1990–2025, with 2000/2008/2011/2015/2018/2020/2022 as crisis episodes), richer feature inventories, and intraday horizons are the concrete next test, and the protocol pre-commits its statistics (Section 5.10).

2. **Small real universes.** Five crypto pairs and three commodities at daily frequency; the E0 null is bounded to these panels, these labelers (L1/L2), and this feature inventory. A failed gate on richer data would be informative; a passed gate would activate the E1 protocol on real data. Neither has been run.

3. **Assumption 1 idealizes episodes.** Real episodes are neither independent (macro memory links crises) nor exchangeable (secular drift in market microstructure), and crisis regimes offer 5–7 episodes at best—the $O(B/\sqrt{E_r})$ estimation terms in Lemma 1 are not decorative. The sub-episode and cross-asset mitigations of Section 3.3 enlarge the environment set but introduce their own dependence; we flag both rather than resolve them.

4. **Theory register.** Propositions 1–3 are idealized-model results: the reward-driven eviction rate assumes per-step foreign-update probability $p > 0$ (verified by construction in our arms, plausible but unverified for arbitrary allocators), and we claim no adversarial-regret guarantee for the coupled winner-take-all dynamics. The bound's additivity motivates but does not entail the empirical super-additivity; we measure the latter.
5. **The mechanism inverts under strong exploitable drift.** E6 is a limitation stated as an experiment: at high within-regime SNR, invariant retention loses to episode-chasing ERM by up to 49%. Any deployment must locate itself on the E6 axis first; a staleness trigger in the GAUSE style (re-opening competition on confident underperformance) is the known remedy and is future work here.
6. **Capacity as a binding constraint is a modeling choice.** As in GAUSE, a purely tabular learner could hold all regimes; the contribution is conditional on capacity binding (true for fixed-size parametric models and for the operational reading—calibration attention, infrastructure—of strategy pools). The soft-interference results (E2) show the dissociation does not depend on the discrete eviction model.
7. **Task-incremental scope.** Every arm receives the same causal regime label; the comparison is information-symmetric, but inferring regimes while specializing (class-incremental) is unsolved here. GAUSE's label-free variant collapsed under real feature overlap, and we expect the financial case to be at least as hard; we do not claim otherwise.

8 Conclusion

We decomposed non-stationary financial decision error into two independently controlled failure modes—between-regime forgetting, owned by the capacity allocator, and within-regime drift, owned by the training objective—and showed theoretically and experimentally that fixing either without the other forfeits most of the available improvement, while fixing both recovers the hand-designed skyline without hand design. The enabling property, imported from GAUSE and ported to decision-focused learning, is reward-independence of the converged assignment; the new axis, imported from invariant predict-and-optimize and ported to recurring regimes, is episode-variance penalization of a decision-focused loss; the bridge between their guarantee languages is one Pinsker step. Around the mechanism we built the evaluation standard we believe the area needs: pre-registered orderings, a structure diagnostic that runs before claims, audits that locate every boundary of validity, and the discipline of reporting the predictions the data refused. When the regime returns, the specialist who kept the playbook—and kept it general—decides best; this paper says precisely when that sentence is true, and how to check.

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